

A Novel Approach to Implement a 4-Bit Vedic Multiplier Using Reversible Techniques

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Abstract—Reversible logic is pivotal in the realm of low-power computing as it eliminates power dissipation associated with information loss, a critical factor in modern electronics where energy efficiency is paramount. The importance of reversible logic extends to quantum computing, where reversible operations are essential for the construction of quantum gates and circuits. Adders play a crucial role in multipliers, especially in digital signal processing (DSP) applications. Vedic multiplication using the Urdhva Triyakbham method is an ancient Indian technique that performs multiplication by vertically and crosswise combining digits. In this research work, the design and implementation of a 4x4 reversible gate known as an ASK gate is presented. This gate is subsequently utilised in the construction of a full adder as well as a 4-bit Vedic multiplier. In terms of factors such as garbage outputs(GO), ancilla input(AI), and the number of gates (GC) that are utilised, the design that was developed performs better than earlier designs.

Index Terms—Reversible logic, Ancilla Inputs, Garbage outputs, Low power, Quantum Computing, Vedic multiplier.

I. INTRODUCTION

The necessity for energy-efficient computing has contributed to increased interest in reversible logic approach. Reversible gates, unlike ordinary logic gates, provide no data loss during computation, leading in zero power usage [1]. This characteristic is particularly advantageous for the development of low-power circuits, which are crucial for a variety of modern applications including quantum computing, optical computing, and low-power VLSI design. This attribute is particularly advantageous in quantum computing, where maintaining coherence and minimizing energy loss are crucial for efficient computation. Reversible logic gates, by ensuring that each input state can be uniquely recovered from the output state, lay the groundwork for constructing quantum circuits that are not only energy-efficient but also capable of handling complex computational tasks [2].

The significance of reversible logic extends beyond quantum computing into fields like nanotechnology and optical computing. In these domains, the ability to minimize heat

generation and energy consumption is paramount, especially as devices continue to shrink in size. Reversible gates enable the development of circuits that produce minimal heat, thereby enhancing the longevity and reliability of nanoscale devices. Additionally, optical computing, which relies on light for data transmission and processing, benefits from the low-energy dissipation properties of reversible gates, leading to faster and more efficient optical circuits. One of the primary advantages of reversible logic gates is their application in designing efficient arithmetic circuits, such as adders and multipliers, which are fundamental components in digital signal processing (DSP) and various computational systems. Traditional adder circuits often generate excessive garbage outputs—unnecessary intermediate data that do not contribute to the final result [3].

Reversible logic gates also adhere to specific properties: they do not allow fan-out, where a gate's output is used as input to multiple gates, and they often require additional constant inputs and produce extra "garbage" outputs to maintain reversibility. Applications of reversible logic are vast, ranging from quantum computing, where many quantum gates are naturally reversible, to low-power CMOS design and cryptography. Multipliers are fundamental components in digital signal processing (DSP) and various arithmetic operations. Traditional multipliers, however, are power-hungry and contribute to significant energy consumption in digital systems. The advent of Vedic mathematics, with its efficient algorithms for multiplication, provides an opportunity to design multipliers that are both fast and energy-efficient. When integrated with reversible logic, Vedic multipliers can further enhance computational efficiency by minimizing power dissipation [4-7].

Urdhva-Triyakbham, also known as the Vertically and Crosswise Sutra, is an example of such an algorithm that originates from Vedic mathematics. This particular algorithm is well-known for its ease of use and effectiveness in the process of conducting multiplication. The Urdhva-Triyakbham

sutra breaks down the multiplication process into smaller, more manageable parts by performing operations both vertically and crosswise. This method not only speeds up the multiplication process but also simplifies the implementation of complex multipliers, making it an excellent candidate for integration with reversible logic [8-9].

This paper introduces a novel 4x4 reversible gate tailored for constructing efficient arithmetic circuits. The proposed gate is utilized to design a full adder, a critical building block in digital circuits, and a 4-bit Vedic multiplier. By focusing on reducing garbage outputs and minimizing the number of ancilla inputs, the proposed designs demonstrate marked improvements over existing solutions. The full adder and 4-bit Vedic multiplier designed using the proposed gate not only maintain functional correctness but also achieve better performance metrics, leading to more efficient and sustainable digital systems. Consider two 2-bit values, $A(A_1, A_0)$ and $B(B_1, B_0)$, for 2×2 multiplication. One of the first steps in the procedure is multiplying the least significant bits (LSBs) of both A (A_0) and B (B_0). ($s_0 = A_0B_0$) is the value that is saved for the outcome. After that, we multiply A_0 by the next bit in B_1 and vice versa ($A_1 \times B_0$ and vice versa). These products are added together and stored as c_1s_1 ($c_1s_1 = A_1B_0 + A_0B_1$). The carry bit (c_1) from the previous step is added to the products of multiplying the remaining bits i.e ($A_1 \times B_1$). In Fig. 1, we can see the 2×2 multiplication and Fig. 2 depicts the schematic for a 2-bit vedic multiplier that makes use of half adders (HA).

Reversible logic gates rely on specific properties to function within reversible computing systems [13]. Here's a breakdown of some key parameters:

Quantum Cost(QC):Quantum cost refers to the total resources required to implement a reversible gate in a quantum computing system. This includes the number of basic quantum operations (such as quantum gates) needed, the depth of the circuit, and the complexity of the gates used.

Garbage Outputs(GO):Garbage outputs are the additional output lines produced by a reversible gate that do not directly contribute to the intended computation but are necessary to ensure the reversibility of the gate. Despite the fact that these outputs are not a part of the usable output, they are necessary in order to maintain the one-to-one mapping of input to output in reversible logic computing.

Gate Count(GC):This refers to the number of basic building block gates (either 1-input, 1-output or 2-input, 2-output) required to create the entire circuit.

Ancillae Inputs(AI):Ancillae (or Ancilla Inputs) refer to the additional input bits required in a reversible logic circuit to assist in the computation. These extra inputs are necessary for the construction of the reversible design, often used to facilitate certain operations while maintaining the reversibility of the circuit.

The proposed research work is organized as follows: Section I lays the foundation with an introduction to reversible logic gates. Section II explores relevant research in the field to provide context. Section III details the design of the proposed

4x4 reversible gate and its application in constructing a full adder and a 4 bit Vedic multiplier. Section IV provides a comparative analysis of Vedic multiplier and full adder with existing architectures. Section V constitutes the Conclusion of the proposed research work and offers insights into future research directions.

II. RELATED WORKS

In order to achieve the best performance in a low-power mode, Sujata S. Chiwande et al., [10] designed a full adder using reversible logic architecture. One of the characteristics that is typical of very large scale integration (VLSI) systems is reversible logic, which contributes to the optimisation of efficiency. This study presents an overview and comparison of two different types of full adder circuits: a conventional full adder circuit and a reversible full adder circuit that utilises the use of a Peres gate. When comparison with the conventional adder circuit, the overall thermal power dissipation features a 44% improvement. Sarangam K et al., [11] proposed the method of opts for low-power architectures, including quantum computing and the Internet of Things. Typical combinational logic circuits are inefficient and consume energy. As reversible computing retains the input bits in the output, this becomes less of contention. An improved low-power reversible full adder for energy-efficient applications is presented in this paper. It is advised that traditional CMOS circuits be compared more closely with energy-efficient full adders in reversible circuits. Commonly used measurements for energy dissipation in complementary metal-oxide semiconductor (CMOS) circuits are $0.5CV_{dd}^2$, which is similar to $0.5CV_{Thr}^2$ in reversible logic. The amount of energy lost is equal to $(V_{Thr}/V_{dd})^2$. In addition, the size of ancillae and trash outputs has been reduced by 16.67%. There are also 16.67% fewer transistors than before.

For the reversible implementation of a 4-bit Vedic multiplier circuit, Vandana Shukla et al. [12] detailed a novel approach with optimal performance characteristics. The fundamental principles of Vedic mathematics form the basis for Vedic multipliers. The multiplier is very efficient since it just takes one step to compute the sum of all the partial products. Implementing this multiplier using a reversible approach will also lead to the construction of digital system multiplier circuits that are efficient, fast, and loss-less. Several criteria are studied in order to determine the circuit's performance. These include the quantum cost (QC), the number of gates (TC), the constant inputs (CI), the garbage outputs (GO), and the suggested multiplier design. Applications requiring low power consumption and high performance utilise the benefits of the 4-bit reversible vedic multiplier introduced by R Saligram et al., [13]. One method for evaluating the aforementioned architecture is the Total Reversible Logic Implementation Cost, or TRLIC for acronym. Using this approach yields a 33% TRLIC compared to related literature works.

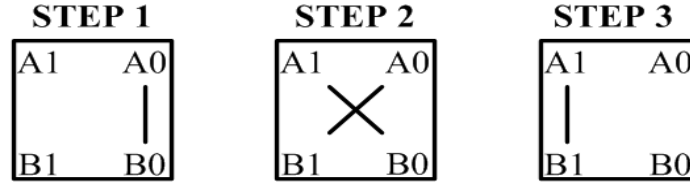


Fig. 1. Implementation of 2-bit Vedic multiplication process using Urdhva Triyakbhyam Principle

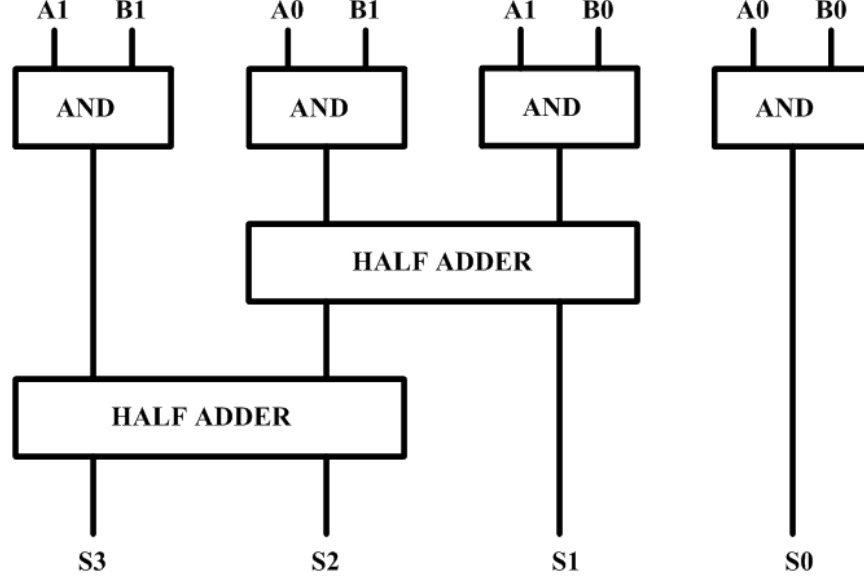


Fig. 2. Designing of 2-bit Vedic multiplier (2-bit VM) using two Half adders(HA)

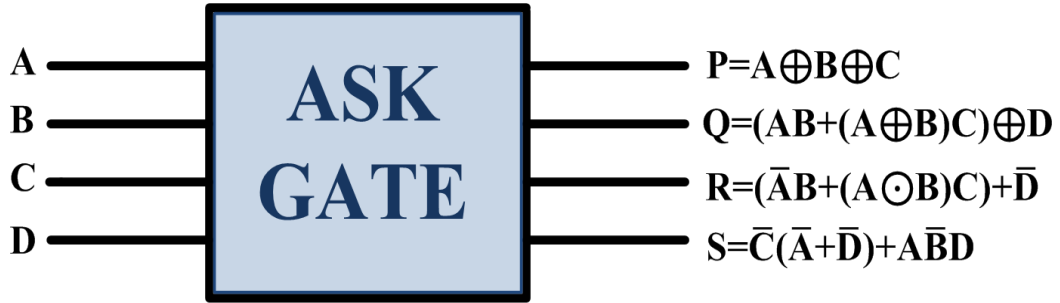


Fig. 3. Proposed reversible ASK Gate

III. PROPOSED DESIGN

This work presents a novel reversible logic gate, designated as the ASK Gate as illustrated in Fig. 3. A key strength of this gate lies in its inherent capability to operate independently as a reversible full adder, as demonstrated in Fig. 4. By setting the input SEL to zero, the design exhibits full adder behavior. This paradigm shift in design methodology condenses the realization of full adders with two garbage outputs into a

single, unified gate, demonstrably yielding performance improvements over prior architectures.

A. Implementation of 4-Bit Vedic Multiplier (4 x 4 VM) using proposed ASK reversible gate

The term "4-bit Vedic multiplier" refers to a digital circuit that is meant to execute multiplication via the Vedic mathematics approach, which is especially developed for binary values with four bits. In comparison to the conventional methods of

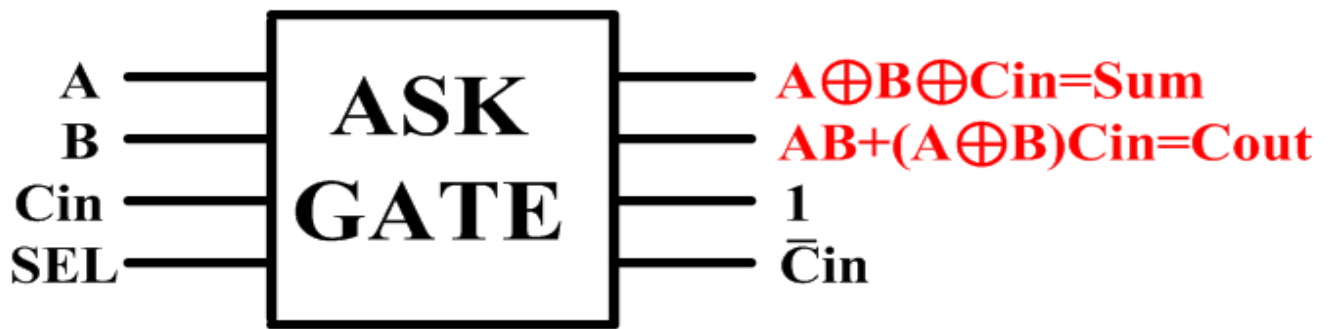


Fig. 4. Implementation of Full adder using proposed ASK Gate

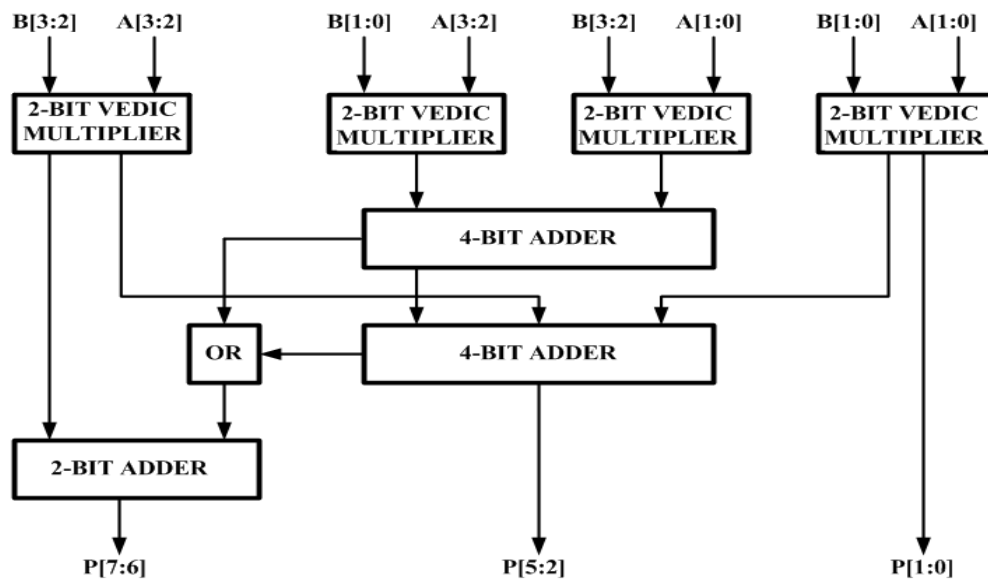


Fig. 5. Implementation of 4-bit Vedic multiplier(VM) using proposed ASK Gate

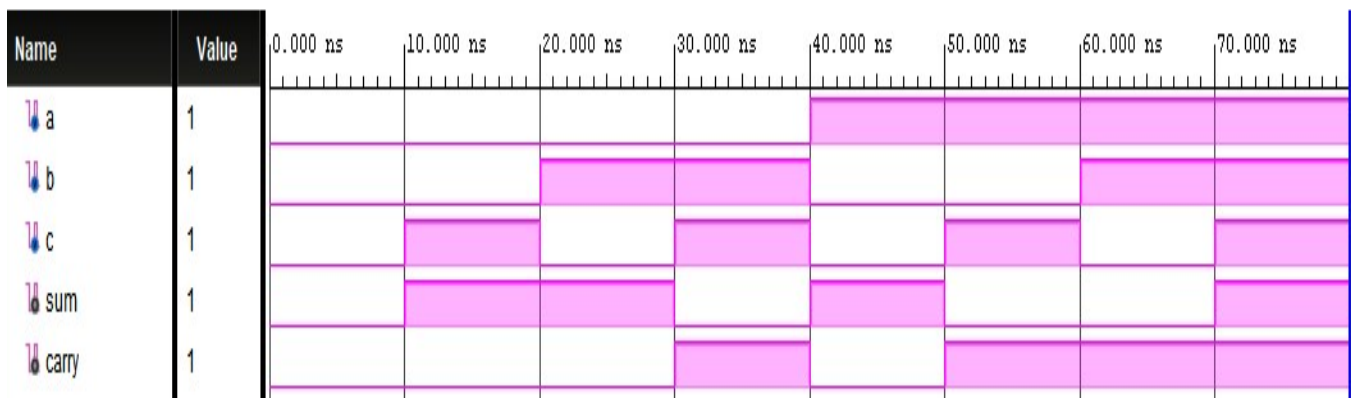


Fig. 6. Simulation output of full adder using proposed ASK Gate

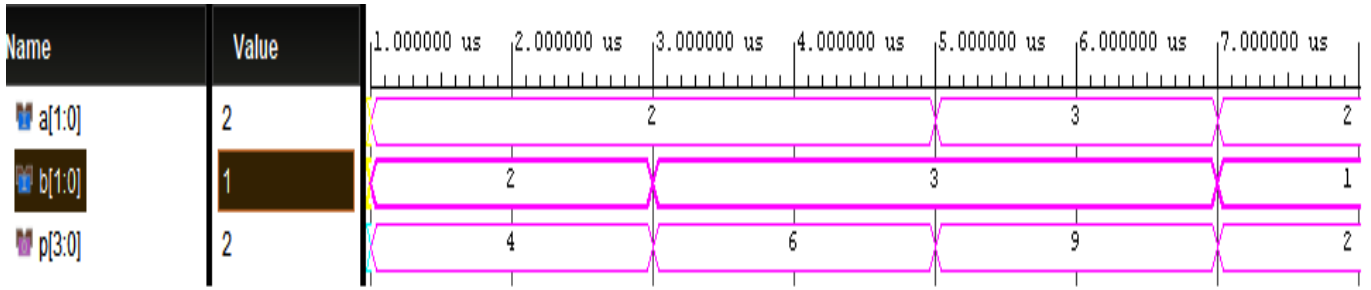


Fig. 7. Simulation output of 2-bit Vedic multiplier

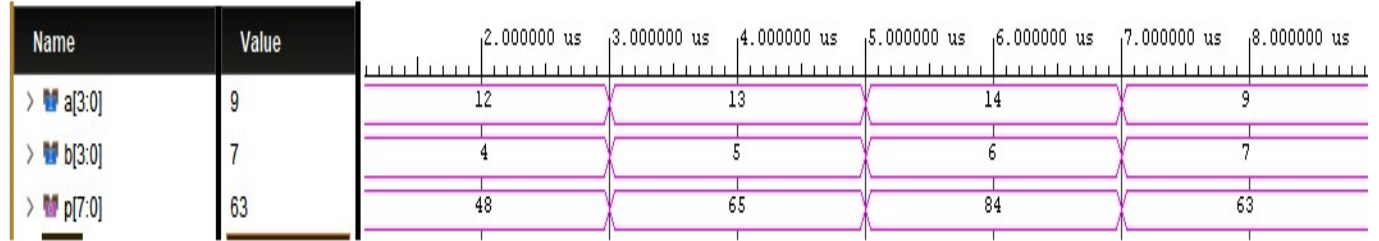


Fig. 8. Simulation output of 4-bit Vedic multiplier using proposed ASK Gate

TABLE I
COMPARISON OF PROPOSED 4-BIT VEDIC MULTIPLIER USING REVERSIBLE ASK GATE WITH OTHER VEDIC MULTIPLIER ARCHITECTURES

Architectures	Ancilla	Garbage Outputs (GO)	Gate Count(GC)
Ref [18]	55	56	64
Ref [19]	58	58	53
Ref [20]	52	52	52
Ref [21]	52	52	52
Proposed	29	50	31

multiplication, the Vedic multiplication methodology, which was derived from ancient Indian mathematics, provides a variant approach. Reversible logic gates are used to build the four 2×2 Vedic multipliers that are included in the design of the 4×4 Vedic multiplier. The design also includes two ripple carry adders with four bits each. Using the Urdhva-Tiryagbhyam sutra, the building of a four-by-four multiplier is discussed in Fig. 5. The first partial product is obtained by multiplying the lowest two bits of a ($a[1:0]$) with the lower two bits of b ($b[1:0]$). This results in the first imperfect product. Multiplying $a[3:2]$ and $b[1:0]$ creates the second partial product that is created. The third partial product is that which is obtained by multiplying $a[1:0]$ and $b[3:2]$. Multiplication is the fourth partial product that may be obtained by multiplying $a[3:2]$ and $b[3:2]$. The second and third partial products, which are presumed to have the same amount of bits, are combined by means of a complete adder that operates on four bits in this architecture. This adder's output is then mixed with the output of another adder that is also four bits in size. The two least significant bits (LSBs) from the fourth partial product and the two most significant bits (MSBs) from the first partial product

are supplied as input1 to the second adder. Additionally, the sum output of the first adder is used to derive an additional input. By the end of the process, a 2-bit adder combines the carry that was generated by the second 4-bit adder with the two most significant bits of the fourth partial product.

IV. RESULTS AND DISCUSSIONS

The design of a novel gate that employs reversible logic gate technology was assessed using Vivado 2022.2 software, with the source code written in Verilog HDL. An ASK gate with $D=0$ set as the input is used in the design of a complete adder. Fig. 6 shows the results of the simulation of the complete adder circuit using the reversible gate that has been presented. The whole adder's capabilities is shown by the various input combinations and their corresponding outputs [14–17]. The inputs are added binary in the Sum output, and any carry-out from the addition operation is shown in the Cout output. Fig. 7 depicts the waveform of a 2-bit Vedic multiplier. In Fig. 8, we can see the waveform of a 4-bit Vedic multiplier that was developed with the reversible gate integrated.

Table I presents a comparison of the number of reversible gates required, the amount of garbage outputs (GO) generated, and the ancilla inputs for the 4-bit Vedic multiplier designed using ASK gate against existing architectures. The proposed circuit uses 29 ancilla bits, employs a total of 31 reversible gates, and produces 50 garbage outputs.

V. CONCLUSION AND FUTURE WORK

This study emphasises the considerable benefits of reversible logic in low-power computing, especially for energy-efficient applications and in the field of quantum computing, where reversible operations are essential. An effective complete adder and a 4-bit Vedic multiplier based on the Urdhva Triyambakam technique are constructed using the 4x4 ASK reversible gate, which is demonstrated in the proposed design and implementation. The novel approach reduces the amount of garbage outputs, minimises the ancilla bits, and decreases the number of gates required, compared to conventional reversible designs.

Our future aim is to create a double precision floating-point Vedic multiplier for advanced DSP applications. This will be achieved by leveraging the suggested full adder design that incorporates the proposed reversible logic gates.

REFERENCES

- [1] A. Roohi, R. Zand, S. Angizi and R. F. DeMara, A Parity-Preserving Reversible QCA Gate with Self-Checking Cascadable Resiliency, in IEEE Transactions on Emerging Topics in Computing, vol. 6, no. 4, pp. 450-459, 1 Oct.-Dec. 2018.
- [2] S. Raveendran, P. J. Edavoor, N. Y. B. Kumar and M. H. Vasantha, An Approximate Low-Power Lifting Scheme Using Reversible Logic, in IEEE Access, vol. 8, pp. 183367-183377, 2020.
- [3] E. Fredkin and T. Toffoli, Conservative logic, in Collision-based computing. Springer, pp. 47-81, 2002.
- [4] T. Toffoli, "Reversible computing," in Automata, Languages and Programming, ser. Lecture Notes in Computer Science, J. de Bakker and J. van Leeuwen, Eds. Springer Berlin Heidelberg, 1980, vol. 85, pp. 632-644, 1980.
- [5] D. P. Vasudevan, P. K. Lala, J. Di, and J. P. Parkerson, Reversible-logic design with online testability, IEEE transactions on instrumentation and measurement, vol. 55, no. 2, pp. 406-414, 2006.
- [6] H. M. H. Babu, N. Saleheen, L. Jamal, S. M. Sarwar, and T. Sasao, Approach to design a compact reversible low power binary comparator, IET Computers Digital Techniques, vol. 8, no. 3, pp. 129-139, 2014.
- [7] M. Alioto and G. Palumbo, "Modeling of power consumption of adiabatic gates versus fan in and comparison with conventional gates, in International Workshop on Power and Timing Modeling, Optimization and Simulation. Springer, pp. 265-275, 2000.
- [8] Aruru Sai Kumar, T.V.K. Hanumantha Rao, An Adaptive Core Mapping Algorithm on NoC for Future Heterogeneous System-on-Chip, Computers and Electrical Engineering, Volume 95, October 2021.
- [9] Aruru Sai Kumar et al., "Nanosheet Field Effect Transistor Device and Circuit Aspects for Future Technology Nodes," ECS Journal of Solid State Science and Technology, Vol. 12, No. 8, 2023.
- [10] S. S. Chiwande and P. K. Dakhole, "Design and Analysis of Low Power Full Adder using Reversible Logic," 2022 6th International Conference on Electronics, Communication and Aerospace Technology, Coimbatore, India, pp. 146-150, 2022.
- [11] Sarangam K et al., "Design and Energy Dissipation Analysis of Full-Adder for Low Power Applications Using Reversible Logic Gates", 3rd International Conference on Emerging Frontiers in Electrical and Electronic Technologies (ICEFEET- 2023), NIT Patna, 2023.
- [12] Shukla, V., Singh, O. P., Mishra, G. R., & Tiwari, R. K., "A novel approach for reversible realization of 4x 4 bit vedic multiplier circuit," In Advances in VLSI, Communication, and Signal Processing: Select Proceedings of VCAS 2018 (pp. 733-745). Springer Singapore, 2020.
- [13] R. Saligram and T. R. Rakshith, "Optimized reversible vedic multipliers for high speed low power operations," 2013 IEEE Conference on Information & Communication Technologies, Thuckalay, India, pp. 809-814, 2013.
- [14] J. Pujar, S. Raveendran, T. Panigrahi, M. H. Vasantha and N. Kumar Y.B., "Design and Analysis of Energy Efficient Reversible Logic based Full Adder," 2019 IEEE 62nd International Midwest Symposium on Circuits and Systems (MWSCAS), USA, pp. 339-342, 2019.
- [15] M. Aditya, Y. B. N. Kumar and M. H. Vasantha, "Reversible full/half adder with optimum power dissipation," 2016 10th International Conference on Intelligent Systems and Control (ISCO), Coimbatore, India, pp. 1-4, 2016.
- [16] C Ganesh et al., "A Novel Design of Area Efficient Full Adder Architecture using Reversible Logic Gates", 7th International Conference on Devices, Circuits, and Systems(ICDCS), 2024.
- [17] J. Bruce et al., Efficient adder circuits based on a conservative reversible logic gate, in Proceedings IEEE Computer Society Annual Symposium on VLSI, pp. 74-79, 2002.
- [18] Thapliyal, H., M.B. Srinivas and H.R. Arabnia, "A Reversible Version of 4x4 Bit Array Multiplier with Minimum Gates and Garbage Outputs," Int. Conf. Embedded System, Applications (ESA'05), pp: 106-114, 2005.
- [19] H. Thapliyal and M. B. Srinivas, "Novel Reversible Multiplier Architecture Using Reversible TSG Gate," IEEE International Conference on Computer Systems and Applications, pp. 100-103, 2006.
- [20] F. Naderpour and A. Vafaei, "Reversible multipliers: Decreasing the depth of the circuit," 2008 International Conference on Electrical and Computer Engineering, pp. 306-310, 2008.
- [21] M. Ehsanpour, P. Moallem and A. Vafaei, "Design of a novel reversible multiplier circuit using modified full adder," 2010 International Conference On Computer Design and Applications, China, pp. V3-230-V3-234, 2010.